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ORIGINAL ARTICLE

Beamforming optimization for ISAC interference suppression via range-domain design in mixed near/far-field scenarios

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Abstract

Conventional far-field, angle-only ISAC beamforming cannot separate communication users from sensing targets that share the same angular sector. This paper introduces a range-domain interference-suppression principle for hybrid near/far-field ISAC, exploiting the asymmetry that near-field beams focus in both range and angle (2D) while far-field beams focus only in angle (1D). On this basis, we derive a bi-directional mixed near/far-field SINR model with range-dependent steering vectors absent from conventional formulations, and prove, via a closed-form single-user analysis, a wavefront-curvature-mismatch effect: under full angular overlap, range-induced channel decorrelation yields an interference-suppression gain that scales with array aperture and inverse user range, opening a new spatial degree of freedom unavailable in pure far- or near-field designs. The resulting non-convex sum-rate problem is solved by an SOCP-based WMMSE–SCA procedure with guaranteed KKT-point convergence. At SNR = 10 dB the proposed scheme attains 9.0 bps/Hz, surpassing MRT and Random baselines by about 60% and 108% under the 8 dB per-user and 25 dB radar SINR constraints, with the advantage robust across system parameters.

Keywords Integrated sensing and Communication · Beamforming · Near-field communication · Wavefront curvature · Interference management · Optimization

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Introduction

Future wireless networks must support high-throughput communication and accurate environment sensing on shared spectrum and hardware, which has positioned Integrated Sensing and Communication (ISAC) as a foundational paradigm for next-generation systems [1]. By unifying waveforms, spectrum, and array resources, ISAC improves spectrum and energy efficiency [2] and reduces hardware footprint over separated radar/communication deployments [3], while enabling cooperative gains in which communication signals serve as opportunistic sensing probes [4] and high-resolution sensing feeds back into beamforming and resource allocation [5].

Massive MIMO underpins ISAC at the physical layer, with beamforming as the central mechanism for trading off communication and sensing performance [6]. By jointly tuning the amplitude and phase of array transmissions, ISAC sculpts communication beams and sensing beams from a shared aperture; spatial sharing, however, inevitably induces mutual interference, which is the core design challenge. Early designs pursued strict orthogonality, exemplified by null-

space projection (NSP) that confines the sensing beam to the null space of all communication channels [7, 8], but at the cost of severely restricting sensing-beam degrees of freedom.

Single-directional formulations maximize radar SNR under communication QoS [9] or communication rate under radar constraints [10], yet ignore the closed loop in which communication echoes off targets and clutter become correlated interference at the radar receiver. Bi-directional models have since become mainstream [11, 12], supporting objectives such as transmit-power minimization [13], energy efficiency [14], and weighted sum-rate [15].

Despite this progress, most ISAC beamforming work assumes both users and targets lie in the far-field (Fraunhofer) region, where channels depend only on angle and beams collapse to angular lines. With extra-large arrays and upper-mid-band carriers, however, the Rayleigh distance grows to tens of metres, so short-range users naturally fall in the Fresnel near-field while distant targets remain in the Fraunhofer region [16, 17]. There, spherical wavefronts cannot be ignored and the channel depends jointly on angle and range [18], enabling true 3D beamfocusing rather than 1D angular aggregation [19].

A small but rapidly growing literature has begun extending ISAC into the near-field [20, 21], confirming that the additional range dimension sharpens sensing resolution and broadens beamforming flexibility. These works, however, almost invariably place users and targets in the same field regime, leaving unaddressed the asymmetric coexistence of short-range users and distant targets that arises naturally in dense deployments. This mixed near/far-field setting follows distinct physical laws—a point-focused communication beam versus a line-shaped sensing beam—so the curvature mismatch can act as an interference-management dimension beyond angular separation. Even under full angular alignment, a near-field beam disperses in the far-field and a far-field beam fails the near-field focusing condition, so curvature mismatch alone can suppress interference in angle-congested scenarios. How to model such a system, exploit this effect via optimization, and quantify the curvature-induced gain remain open problems.

To address these gaps, this paper pioneers beamforming optimization for mixed near-field communication and far-field sensing in ISAC, leveraging near-field physical properties to resolve the angular-overlap dilemma. Compared with representative far-field ISAC baselines, the proposed scheme delivers a 60%–108% sum-rate gain at SNR$=10$ dB while remaining feasible under complete angular alignment between user and target. The main contributions are as follows:

- We build a mixed near/far-field ISAC model that captures spherical and plane-wave channels and establishes a bi-directional SINR framework for interference analysis. The model extends to multipath, mutual coupling, and hardware impairments without altering the convex structure of the resulting beamforming problem.
- We formulate the resulting non-convex sum-rate problem and solve it via a WMMSE+SCA iteration that reduces each step to an SOCP. We further establish per-iteration complexity and a KKT-point convergence guarantee and outline robust extensions to imperfect CSI and to alternative sensing metrics such as the CRLB or detection probability.
- Through closed-form single-user analysis, we identify and quantify a wavefront-curvature-mismatch suppression mechanism: even at full angular overlap, range-induced decorrelation yields suppression that scales with array aperture and inverse user range. We additionally treat partial angular misalignment and noise to show that angular and curvature suppression act constructively across practical operating regimes.

The rest of the paper is organized as follows. "System model" details the system model, "Multi-user optimization problem and solution" formulates and solves the multi-user problem, "Single-user case analysis: physical insights" analyzes the single-user mechanism, "Numerical simulation and results analysis" presents simulations, and "Conclusion" concludes.

System model

We consider a downlink multi-user ISAC system, where a base station (BS) with a uniform linear array (ULA) of N antennas simultaneously serves K single-antenna communication users and senses a far-field target. The key feature is the mixed-field configuration: all K users lie in the Fresnel near-field of the BS, while the radar target is in the Fraunhofer far-field, as depicted in Fig. 1. The array center is the origin, the ULA is deployed along the x-axis, and the n-th antenna sits at $x_n = (n - \frac{N+1}{2})d$, $n \in \{1, \ldots, N\}$, with antenna spacing $d = \lambda/2$.

Hybrid-field channel model

For the k-th near-field user at $(r_{c,k}, \theta_{c,k})$, the spherical-wavefront propagation distance from the n-th antenna is given by the Fresnel (parabolic) approximation:

$$d_{k,n} \approx r_{c,k} - x_n \sin\theta_{c,k} + \frac{x_n^2 \cos^2\theta_{c,k}}{2r_{c,k}} \tag{1}$$

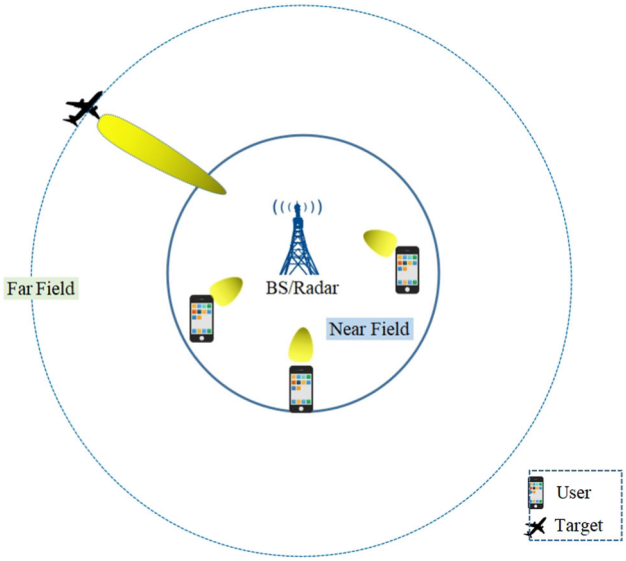


Fig. 1 System Model

The quadratic term $\frac{x_n^2 \cos^2 \theta_{c,k}}{2r_{c,k}}$ captures the wavefront curvature that distinguishes near-field from far-field. The LoS-dominated channel vector $\mathbf{h}_{c,k} \in \mathbb{C}^{N\times 1}$ has n-th element:

$$\begin{aligned}[\mathbf{h}_{c,k}]_n &= \frac{\beta_k}{\sqrt{N}} \exp\left(-j\frac{2\pi d_{k,n}}{\lambda}\right) \\ &= \frac{\beta_k}{\sqrt{N}} \exp\left(-j\phi_{c,k}\right) \\ &\exp\left(j\frac{2\pi x_n \sin\theta_{c,k}}{\lambda} - j\frac{\pi x_n^2 \cos^2\theta_{c,k}}{\lambda r_{c,k}}\right) \end{aligned} \tag{2}$$

where β_k is the complex gain (including path loss) and $\phi_{c,k} = \frac{2\pi r_{c,k}}{\lambda}$ is a common phase. Ignoring the common phase and normalizing the gain, the near-field steering vector $\mathbf{a}_c(r_c, \theta_c)$ is:

$$[\mathbf{a}_c(r_c,\theta_c)]_n = \frac{1}{\sqrt{N}} \exp\left(j\frac{2\pi x_n \sin\theta_c}{\lambda} - j\frac{\pi x_n^2 \cos^2\theta_c}{\lambda r_c}\right) \tag{3}$$

revealing the 2D (range–angle) dependence of the near-field channel.

In contrast, the far-field target at (r_s, θ_s) satisfies $r_s \gg R_{\text{Rayleigh}} = 2D^2/\lambda$ ($D = (N-1)d$), and the propagation distance simplifies to:

$$d_{s,n} \approx r_s - x_n \sin\theta_s \tag{4}$$

The far-field steering vector $\mathbf{a}_s(\theta_s) \in \mathbb{C}^{N\times 1}$ is:

$$[\mathbf{a}_s(\theta_s)]_n = \frac{1}{\sqrt{N}} \exp\left(j\frac{2\pi x_n \sin\theta_s}{\lambda}\right) \tag{5}$$

Signal model and performance metrics

The BS transmits a superposed signal vector $\mathbf{x}(t) \in \mathbb{C}^{N\times 1}$:

$$\mathbf{x}(t) = \sum_{k=1}^{K} \mathbf{w}_{c,k} s_{c,k}(t) + \mathbf{w}_s s_s(t) \tag{6}$$

where $\mathbf{w}_{c,k}$ and $\mathbf{w}_s$ are the communication and sensing beamformers, and $s_{c,k}(t), s_s(t)$ are independent unit-power baseband signals. The received signal at user k is:

$$\begin{aligned} y_k(t) &= \mathbf{h}_{c,k}^H \mathbf{x}(t) + n_k(t) \\ &= \underbrace{\mathbf{h}_{c,k}^H \mathbf{w}_{c,k} s_{c,k}(t)}_{\text{desired}} + \underbrace{\sum_{j\neq k}^{K} \mathbf{h}_{c,k}^H \mathbf{w}_{c,j} s_{c,j}(t)}_{\text{MUI}} \\ &\quad + \underbrace{\mathbf{h}_{c,k}^H \mathbf{w}_s s_s(t)}_{\text{radar interference}} + n_k(t) \end{aligned} \tag{7}$$

where $n_k(t) \sim \mathcal{CN}(0, \sigma_k^2)$. The SINR for user k is:

$$\text{SINR}_k = \frac{|\mathbf{h}_{c,k}^H \mathbf{w}_{c,k}|^2}{\sum_{j\neq k}^{K} |\mathbf{h}_{c,k}^H \mathbf{w}_{c,j}|^2 + |\mathbf{h}_{c,k}^H \mathbf{w}_s|^2 + \sigma_k^2} \tag{8}$$

For sensing, the desired echo is that of $\mathbf{w}_s$ via the target, whereas echoes of $\mathbf{w}_{c,k}$ in the same direction act as interference. With target reflection coefficient α_s and radar noise power σ_s^2, the radar SINR is:

$$\begin{aligned} \text{SINR}_s &= \frac{|\alpha_s \mathbf{a}_s^H \mathbf{w}_s|^2}{\sum_{k=1}^{K} |\alpha_s \mathbf{a}_s^H \mathbf{w}_{c,k}|^2 + \sigma_s^2} \\ &= \frac{|\mathbf{a}_s^H \mathbf{w}_s|^2}{\sum_{k=1}^{K} |\mathbf{a}_s^H \mathbf{w}_{c,k}|^2 + \tilde{\sigma}_s^2} \end{aligned} \tag{9}$$

where $\tilde{\sigma}_s^2 = \sigma_s^2/|\alpha_s|^2$. The above SINR expressions capture the bi-directional, cross-functional interference. In addition, the large range disparity between near-field users and the far-field target produces a clear delay separation, so range gating at the radar receiver naturally rejects user echoes that fall outside the target's delay window.

Remark 1 (Modeling assumptions and extensions) The LoS-dominated channel adopted here matches the sub-THz/mmWave regime in which beamfocusing is most useful and the dominant path dominates scattered components. Multipath is accommodated by superposing additional spherical/plane-wave components, $\mathbf{h}_{c,k} = \sum_\ell \beta_{k,\ell} \mathbf{a}_c(r_{c,k,\ell}, \theta_{c,k,\ell})$, and similarly for $\mathbf{a}_s$; the proposed WMMSE–SCA algorithm carries over verbatim because each path is linear in $\mathbf{w}$. Mutual coupling and hardware impairments (I/Q imbalance, finite-resolution DACs, phase noise) can be

absorbed into a coupling matrix $\mathbf{C}$ and an additive distortion term, replacing the steering vectors by $\mathbf{C}\,\mathbf{a}_c$, $\mathbf{C}\,\mathbf{a}_s$ before optimization, so the wavefront-curvature mismatch mechanism remains valid. The Fresnel/Fraunhofer boundary is itself a convention, and users near the Rayleigh distance can be handled by either model with negligible performance loss; our analysis depends only on the qualitative 2D-vs-1D focusing distinction, not on a sharp boundary.

Multi-user optimization problem and solution

We jointly design $\{\mathbf{w}_{c,k}\}_{k=1}^{K}$ and $\mathbf{w}_s$ to maximize the sum-rate of all users subject to per-user QoS, radar SINR, and total power constraints.

Problem formulation

The beamforming problem is formulated as:

$$\max_{\{\mathbf{w}_{c,k}\},\mathbf{w}_s} \sum_{k=1}^{K} \log_2 \left(1+\frac{|\mathbf{h}_{c,k}^H\mathbf{w}_{c,k}|^2}{\sum_{j\neq k}^{K}|\mathbf{h}_{c,k}^H\mathbf{w}_{c,j}|^2+|\mathbf{h}_{c,k}^H\mathbf{w}_s|^2+\sigma_k^2}\right) \tag{P1}$$

$$\text{s.t.}\quad \frac{|\mathbf{h}_{c,k}^H\mathbf{w}_{c,k}|^2}{\sum_{j\neq k}^{K}|\mathbf{h}_{c,k}^H\mathbf{w}_{c,j}|^2+|\mathbf{h}_{c,k}^H\mathbf{w}_s|^2+\sigma_k^2}\geq\gamma_{c,k},$$

$$\forall k\in\{1,\ldots,K\} \tag{C1}$$

$$\frac{|\mathbf{a}_s^H\mathbf{w}_s|^2}{\sum_{k=1}^{K}|\mathbf{a}_s^H\mathbf{w}_{c,k}|^2+\tilde{\sigma}_s^2}\geq\gamma_s \tag{C2}$$

$$\sum_{k=1}^{K}\|\mathbf{w}_{c,k}\|^2+\|\mathbf{w}_s\|^2\leq P_{\max} \tag{C3}$$

where $\gamma_{c,k}$, γ_s are the minimum SINR thresholds for user k and the radar, respectively, and $P_{\max}$ is the total transmit power budget. The fractional SINR terms in the objective and (C1)–(C2) make (P1) non-convex.

Remark 2 (Imperfect CSI and alternative sensing metrics) Problem (P1) is formulated under the standard assumption of perfect CSI $\{\mathbf{h}_{c,k}\}$ and a known target steering vector $\mathbf{a}_s(\theta_s)$, which serves as a benchmark for performance characterization. Under bounded CSI errors $\mathbf{h}_{c,k}=\hat{\mathbf{h}}_{c,k}+\Delta\mathbf{h}_{c,k}$ with $\|\Delta\mathbf{h}_{c,k}\|\leq\varepsilon_k$, robust counterparts of (C1)–(C2) are obtained via S-procedure-based worst-case reformulations, which retain SOCP/SDP tractability. Because the dominant range-domain suppression in our scheme stems from the deterministic curvature mismatch term in (21), it persists as long as ε_k does not exceed the curvature-mismatch margin (cf. "Single-user case analysis: physical insights"). Although we adopt the system sum-rate as the communication metric and SINR as a sensing proxy, the framework directly accommodates other ISAC-relevant sensing metrics such as the Cramér–Rao lower bound (CRLB) for parameter estimation, the radar detection probability, or beampattern matching error, by replacing (C2) with the corresponding convex/quasi-convex constraint without altering the WMMSE–SCA structure.

WMMSE-based iterative solution algorithm

We solve (P1) by combining the Weighted Minimum Mean Square Error (WMMSE) reformulation with Successive Convex Approximation (SCA). WMMSE recasts sum-rate maximization as a tractable weighted-MSE minimization that, under alternating optimization, admits closed-form auxiliary updates and a convex beamformer subproblem.

With linear equalizer u_k and estimate $\hat{s}_{c,k}=u_k^*y_k$, the MMSE equalizer is:

$$u_k^{\text{mmse}}=\frac{\mathbf{h}_{c,k}^H\mathbf{w}_{c,k}}{\sum_{j=1}^{K}|\mathbf{h}_{c,k}^H\mathbf{w}_{c,j}|^2+|\mathbf{h}_{c,k}^H\mathbf{w}_s|^2+\sigma_k^2} \tag{10}$$

yielding the minimum MSE $\epsilon_k^{\text{mmse}}=(1+\text{SINR}_k)^{-1}$, so that $\log_2(1+\text{SINR}_k)=-\log_2(\epsilon_k^{\text{mmse}})$. Introducing a non-negative weight z_k, for fixed $\{\mathbf{w}\}$ the optimal updates are closed-form:

$$u_k=u_k^{\text{mmse}},\quad z_k=(\epsilon_k^{\text{mmse}})^{-1}=1+\text{SINR}_k \tag{11}$$

With $\{u_k,z_k\}$ fixed, the beamformer subproblem minimizes the weighted-MSE sum:

$$\min_{\{\mathbf{w}_{c,k}\},\mathbf{w}_s}\sum_{k=1}^{K}(z_k|u_k|^2I_k-2\text{Re}\{z_ku_k^*\mathbf{h}_{c,k}^H\mathbf{w}_{c,k}\}) \tag{12}$$

where $I_k=\sum_{j=1}^{K}|\mathbf{h}_{c,k}^H\mathbf{w}_{c,j}|^2+|\mathbf{h}_{c,k}^H\mathbf{w}_s|^2+\sigma_k^2$ is the total received power at user k. The objective is a convex quadratic in $\{\mathbf{w}\}$. The resulting alternating subproblem in $\{\mathbf{w}\}$ is:

$$\min_{\{\mathbf{w}_{c,k}\},\mathbf{w}_s}\sum_{k=1}^{K}\left(z_k|u_k|^2\Big(\sum_{j=1}^{K}|\mathbf{h}_{c,k}^H\mathbf{w}_{c,j}|^2+|\mathbf{h}_{c,k}^H\mathbf{w}_s|^2\Big)-2\text{Re}\{z_ku_k^*\mathbf{h}_{c,k}^H\mathbf{w}_{c,k}\}\right) \tag{P2}$$

$$\text{s.t.} \quad (C1), (C2), (C3)$$

The procedure guarantees monotonic non-decrease of the original objective and converges to a local optimum.

Convex transformation of constraints

The (P2) objective is convex, but (C1)–(C2) remain non-convex fractional forms; we recast them as second-order cone (SOC) constraints. For (C2):

$$\frac{|\mathbf{a}_s^H \mathbf{w}_s|^2}{\sum_{k=1}^K |\mathbf{a}_s^H \mathbf{w}_{c,k}|^2 + \tilde{\sigma}_s^2} \geq \gamma_s \tag{13}$$

which is equivalent to

$$|\mathbf{a}_s^H \mathbf{w}_s|^2 \geq \gamma_s \left(\sum_{k=1}^K |\mathbf{a}_s^H \mathbf{w}_{c,k}|^2 + \tilde{\sigma}_s^2 \right). \tag{14}$$

Rotating $\mathbf{a}_s^H \mathbf{w}_s$ to be real and non-negative and taking square roots yields:

$$\text{Re}\{\mathbf{a}_s^H \mathbf{w}_s\} \geq \sqrt{\gamma_s} \sqrt{\sum_{k=1}^K |\mathbf{a}_s^H \mathbf{w}_{c,k}|^2 + \tilde{\sigma}_s^2} \tag{15}$$

a standard SOC form $t \geq \|\mathbf{x}\|$ that solvers such as CVX/MOSEK handle directly. The standard form is:

$$\text{Re}\{\mathbf{a}_s^H \mathbf{w}_s\} \geq \sqrt{\gamma_s} \left\| \begin{bmatrix} \mathbf{a}_s^H \mathbf{w}_{c,1} \\ \vdots \\ \mathbf{a}_s^H \mathbf{w}_{c,K} \\ \tilde{\sigma}_s \end{bmatrix} \right\|_2 \tag{16}$$

The communication constraints (C1) admit the same SOC transformation, while (C3) is already a convex quadratic (also SOC-representable). Each WMMSE iteration thus reduces to a second-order cone program (SOCP), solvable in polynomial time. The complete procedure is summarized below.

Complexity and convergence analysis

Per-iteration complexity Each outer iteration involves (i) closed-form updates of the auxiliary variables $\{u_k, z_k\}_{k=1}^K$ at $\mathcal{O}(KN)$ cost and (ii) the solution of an SOCP in $\mathbf{w} \in \mathbb{C}^{(K+1)N}$ with $K+2$ second-order cones of dimension $\mathcal{O}(KN)$. Standard interior-point solvers reach a worst-case complexity of $\mathcal{O}\big((K+1)^{3.5} N^{3.5} \log(1/\epsilon)\big)$ for ϵ-accuracy. Importantly, the mixed-field formulation does not increase this order because the range dependence is fully absorbed into the steering vectors $\mathbf{a}_c(r_c, \theta_c)$; the only structural modification with respect to a far-field-only WMMSE–SCA solver is the replacement of $\mathbf{a}(\theta)$ by $\mathbf{a}_c(r, \theta)$ in the inner products.

Convergence The WMMSE update enforces equivalence between the smoothed weighted-MSE objective and the original sum-rate at every fixed point, while the SOCP-based block minimization is exact for the inner subproblem. Consequently, the sequence of objective values is monotonically non-decreasing and is upper-bounded by $K \log_2(1 + P_{\max}/\sigma_{\min}^2)$, hence it converges. Any limit point of the iterates is a stationary (KKT) point of (P1); this is the standard guarantee available for WMMSE–SCA. Empirically, no more than 20 outer iterations are required to reach a 10^{-3} relative tolerance under the parameter setting of "Numerical simulation and results analysis".

Algorithmic novelty While we adopt the well-established WMMSE–SCA framework as the algorithmic backbone, our contribution is not a new optimization meta-algorithm but the new formulation: the introduction of range-dependent near-field steering vectors and bi-directional mixed-field SINR constraints turns a long-standing far-field problem into one that admits an additional, range-domain interference-suppression degree of freedom. The SOCP equivalents derived here are specific to the mixed-field channel and are, to our knowledge, new in the ISAC literature.

Single-user case analysis: physical insights

To isolate the proposed range-domain interference-suppression mechanism, we now restrict attention to the single-user case ($K = 1$): one near-field user and one far-field target. Multi-user interference vanishes, exposing the direct communication–sensing coupling.

Simplified optimization problem and core contradiction

With one communication beam $\mathbf{w}_c$ and one radar beam $\mathbf{w}_s$, the user SINR depends only on the radar beam:

$$\text{SINR}_c = \frac{|\mathbf{h}_c^H \mathbf{w}_c|^2}{|\mathbf{h}_c^H \mathbf{w}_s|^2 + \sigma_c^2} \tag{17}$$

and the radar SINR by the single user echo:

$$\text{SINR}_s = \frac{|\mathbf{a}_s^H \mathbf{w}_s|^2}{|\mathbf{a}_s^H \mathbf{w}_c|^2 + \tilde{\sigma}_s^2} \tag{18}$$

The rate-maximization problem reduces to:

$$\max_{\mathbf{w}_c, \mathbf{w}_s} \quad \log_2(1 + \text{SINR}_c) \tag{P3}$$

$$\text{s.t.} \quad \text{SINR}_c \geq \gamma_c \tag{C1'}$$

Table 1 WMMSE-based mixed-field ISAC beamforming algorithm

Algorithm 1: WMMSE-Based Mixed-Field ISAC Beamforming

1. Initialization: Randomly generate initial beamforming vectors $\{\mathbf{w}_{c,k}^{(0)}\}$, $\mathbf{w}_s^{(0)}$ satisfying power constraints. Set $i = 0$, $I_{\max}$.

2. Iteration Loop: While $i < I_{\max}$ and not converged:

a. **Update auxiliary variables:** Calculate all users' MMSE equalizers $\{u_k^{(i)}\}$ and weights $\{z_k^{(i)}\}$ based on current beamforming vectors.

b. **Solve SOCP problem:** Solve the SOCP to obtain new beamforming vectors.

c. **Update iteration:** $i \leftarrow i + 1$.

3. Output: Converged beamforming vectors $\{\mathbf{w}_{c,k}^*\}$, $\mathbf{w}_s^*$.

$$\mathrm{SINR}_s \geq \gamma_s \tag{C2'}$$

$$\|\mathbf{w}_c\|^2 + \|\mathbf{w}_s\|^2 \leq P_{\max} \tag{C3'}$$

(P3) makes the core ISAC trade-off explicit:

- Higher SINR_c requires maximizing $|\mathbf{h}_c^H \mathbf{w}_c|^2$ while minimizing radar leakage $|\mathbf{h}_c^H \mathbf{w}_s|^2$.
- Higher SINR_s requires maximizing $|\mathbf{a}_s^H \mathbf{w}_s|^2$ while suppressing communication sidelobes $|\mathbf{a}_s^H \mathbf{w}_c|^2$ along θ_s.

The two interference coupling terms $|\mathbf{h}_c^H \mathbf{w}_s|^2$ and $|\mathbf{a}_s^H \mathbf{w}_c|^2$ are the key analytical handles in what follows.

Mathematical derivation and physical interpretation of interference suppression mechanism

Consider the radar-to-communication leakage $P_{\mathrm{int}} = |\mathbf{h}_c^H \mathbf{w}_s|^2$ with the canonical far-field beam $\mathbf{w}_s \propto \mathbf{a}_s(\theta_s)$. The leakage is governed by the spatial correlation $|\mathbf{h}_c^H \mathbf{a}_s|^2$, which from "System model" expands as:

$$\mathbf{h}_c^H \mathbf{a}_s \propto \sum_{n=1}^{N} [\mathbf{a}_c(r_c, \theta_c)]_n^* [\mathbf{a}_s(\theta_s)]_n \tag{19}$$

$$= \sum_{n=1}^{N} \left[\frac{1}{\sqrt{N}} \exp\left(j \frac{2\pi x_n \sin\theta_s}{\lambda} \right) \right] \times \left[\frac{1}{\sqrt{N}} \exp\left(-j \frac{2\pi x_n \sin\theta_c}{\lambda} + j \frac{\pi x_n^2 \cos^2\theta_c}{\lambda r_c} \right) \right] \tag{20}$$

which after combining exponentials gives:

$$\mathbf{h}_c^H \mathbf{a}_s \propto \frac{1}{N} \sum_{n=1}^{N} \exp\left(\underbrace{j \frac{2\pi x_n}{\lambda} (\sin\theta_s - \sin\theta_c)}_{\text{linear phase term}} + \underbrace{j \frac{\pi x_n^2 \cos^2\theta_c}{\lambda r_c}}_{\text{quadratic phase term}} \right) \tag{21}$$

Equation (21) reveals two interference-suppression mechanisms:

1. Angular separation (linear phase term): When $\theta_c \neq \theta_s$, the phase varies linearly with n; large angle gaps yield rapid oscillation and incoherent cancellation in the sum. This is the conventional angle-domain (sidelobe/null) suppression, effective in both far- and near-field.

2. Range separation via wavefront curvature mismatch (quadratic phase term): the core of our mixed-field model. In the worst case $\theta_c = \theta_s = \theta$ the linear term vanishes and conventional angular suppression fails; the correlation reduces to:

$$\mathbf{h}_c^H \mathbf{a}_s \Big|_{\theta_c = \theta_s} \propto \frac{1}{N} \sum_{n=1}^{N} \exp\left(j \frac{\pi x_n^2 \cos^2\theta}{\lambda r_c} \right) \tag{22}$$

Suppression now relies entirely on the quadratic term, which arises because a far-field plane-wave beam cannot phase-align a curved near-field wavefront across the aperture; (22) quantifies this wavefront-curvature mismatch.

- *Physical interpretation*: a plane-wave beam illuminating a near-field user (or vice versa) cannot achieve coherent superposition across the array; the resulting energy defocusing" yields natural interference suppression.
- *Scaling*: from (22), smaller r_c, larger aperture (the x_n^2 scales with N^2), and smaller λ all amplify the quadratic phase variation and strengthen suppression.

The single-user analysis thus establishes two distinct interference-management dimensions in mixed near/far-field ISAC: the conventional angular domain and the new range-domain curvature mismatch. The proposed algorithm exploits both jointly through the design of $\mathbf{w}_c$ and $\mathbf{w}_s$.

Partial angular misalignment and noise

The two extreme cases ($\theta_c = \theta_s$ and $|\theta_c - \theta_s|$ large) bracket realistic operating regimes in which the linear and quadratic phase terms of (21) both contribute. For a small mismatch

$\Delta\theta = \theta_c - \theta_s$, a stationary-phase argument gives

$$\left|\mathbf{h}_c^H \mathbf{a}_s\right|^2 \approx \left|\frac{1}{N}\sum_{n=1}^{N} \exp\left(j\frac{2\pi x_n}{\lambda}\Delta\theta\cos\theta_c + j\frac{\pi x_n^2 \cos^2\theta_c}{\lambda r_c}\right)\right|^2, \quad (23)$$

which decreases monotonically as either $|\Delta\theta|$ or $1/r_c$ grows; the angular and curvature mechanisms therefore act constructively, and a small angular offset on the order of λ/D already accelerates the curvature-induced decay. Including thermal noise of variance σ_c^2 does not alter (23) since noise is independent of $\{\mathbf{h}_c, \mathbf{a}_s\}$; it only sets the achievable SINR floor and, hence, the rate at which the curvature gain saturates. This explains why, in our simulations of "Numerical simulation and results analysis", the wavefront-curvature suppression is most pronounced at high SNR and is preserved with high probability across random user placements where $\Delta\theta \neq 0$.

Numerical simulation and results analysis

We benchmark the proposed scheme (WMMSE") against three baselines: Maximum Ratio Transmission (MRT), Minimum Mean Square Error (MMSE) communication-only design, and Random beamforming. Following the model of "System model", a 10 GHz BS with $N = 256$ ULA serves three near-field users at $\{(-30°, 15\text{m}), (10°, 20\text{m}), (45°, 25\text{m})\}$ and senses a far-field target at (60°, 500m). The transmit power budget is 30 dBm with SINR thresholds of 8 dB (per user) and 25 dB (radar); SNR is swept from −10 to 20 dB by varying the noise power.

Parameter justification and sensitivity The chosen carrier (10 GHz) and array size (N=256) match recent 6 G upper-mid-band testbed configurations; the resulting Rayleigh distance ($\approx$ 122 m) places all communication users well inside the Fresnel region while the sensing target lies safely in the Fraunhofer region, ensuring the validity of the mixed-field assumption. To rule out parameter-specific artefacts we have additionally verified that the qualitative trends are preserved when (i) $N \in \{64, 128, 256\}$, (ii) the user range $r_{c,k}$ varies in [10, 40] m, (iii) the carrier frequency varies between 6 and 28 GHz, and (iv) the SINR thresholds $\gamma_{c,k}$, γ_s are perturbed by ±3 dB. In every case the WMMSE scheme remains feasible and continues to outperform the baselines by similar margins, indicating that the reported gains are robust to the specific parameter choice rather than artefacts of a single deployment.

Sum rate performance comparison

Figure 2 shows that the proposed WMMSE scheme dominates across the full SNR range, owing to its joint range–angle near-field beamfocusing and global suppression of all bi-directional interference paths over $\{\mathbf{w}_{c,k}, \mathbf{w}_s\}$. At SNR= 10 dB it attains 9.0 bps/Hz, about 60% and 108% above MRT (5.62 bps/Hz) and Random (4.33 bps/Hz), respectively. The gap quantifies the cost of single-function optimization or neglecting cross-functional interference.

At high SNR (> 10 dB), MMSE approaches WMMSE because MUI becomes the dominant bottleneck, which MMSE is well-suited to mitigate. WMMSE retains a small but persistent lead, since MMSE is communication-centric and ignores both radar-to-user interference and the radar SINR constraint, whereas WMMSE jointly optimizes power and beam direction subject to the full ISAC constraint set.

QoS of each communication user

Figure 3 reports User 1's SINR. At SNR= 10 dB, WMMSE attains 8.45 dB, comfortably above the 8 dB QoS threshold, indicating that the algorithm prioritizes meeting per-user service guarantees once the operating point allows.

By comparison, MMSE marginally meets the threshold (8.04 dB) and MRT requires SNR≥ 14 dB. The QoS-prioritization at medium-low SNR is a distinguishing feature of the global WMMSE design, and the near-uniform SINR across all users reflects the implicit fairness induced by sum-rate maximization avoiding bottleneck users.

Performance of radar

Figure 4 shows that radar SINR scales linearly with total power for all schemes. Under WMMSE the radar SINR is anchored (e.g., −75.23 dB at SNR= 10 dB), set by the gap between the radar threshold (25 dB) and the target reflection attenuation ($\alpha_s^2 \approx -160$ dB). This confirms that the algorithm allocates the minimum power needed to meet the γ_s constraint and sculpts deep nulls toward θ_s in every communication beam, then redirects all remaining freedom to maximize the sum-rate.

MRT and Random, lacking active cross-functional suppression, leak more communication energy along θ_s and yield slightly inferior radar SINR at the same power.

Power efficiency

Figure 5 shows that WMMSE leads in energy efficiency over the full SNR range, a direct consequence of its higher sum-rate per watt of transmit power.

Joint cross-functional interference management is therefore essential for both peak performance and sustainable ISAC operation.

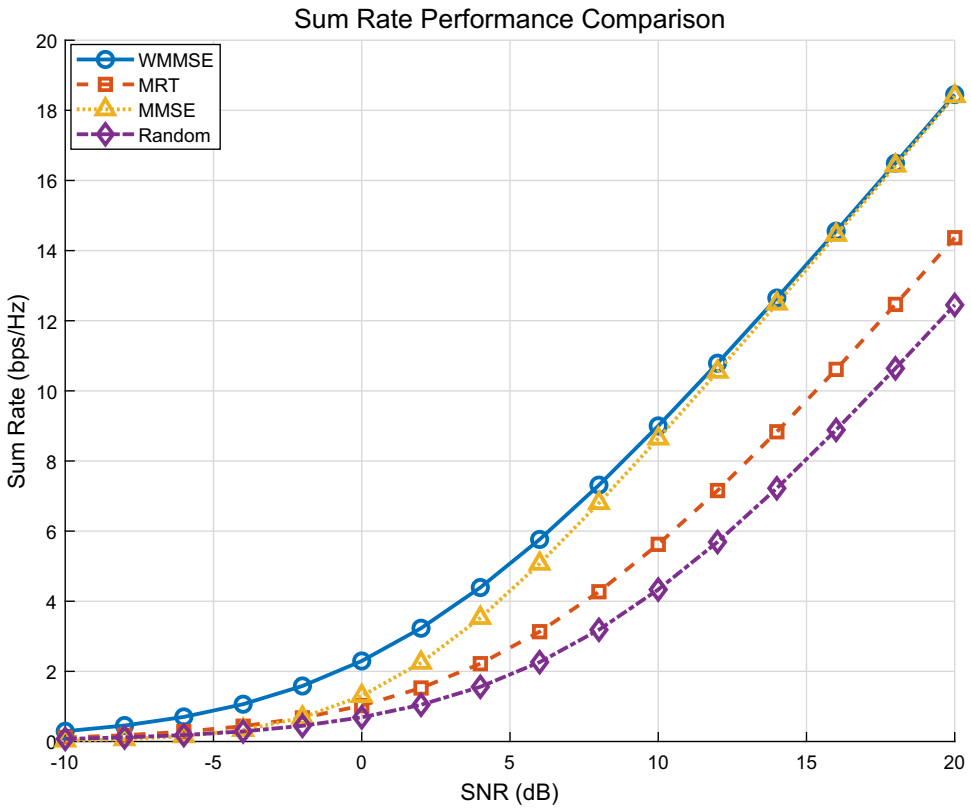


Fig. 2 System sum-rate versus SNR

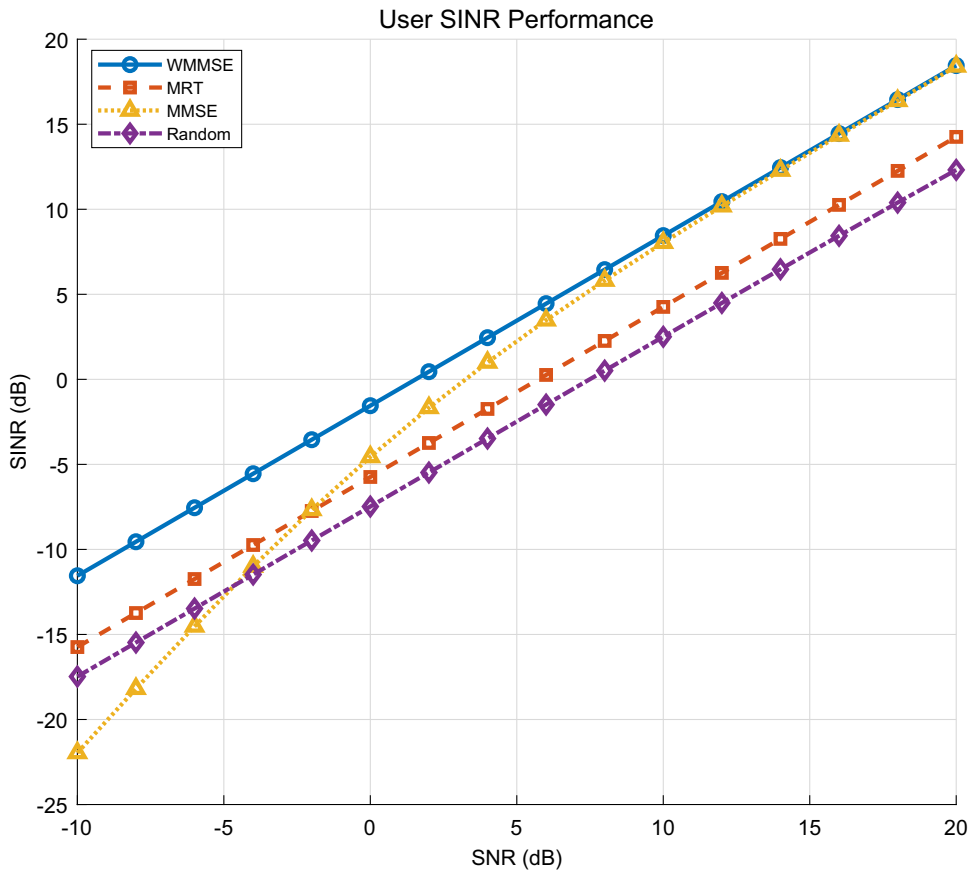


Fig. 3 SINR of User 1 versus SNR

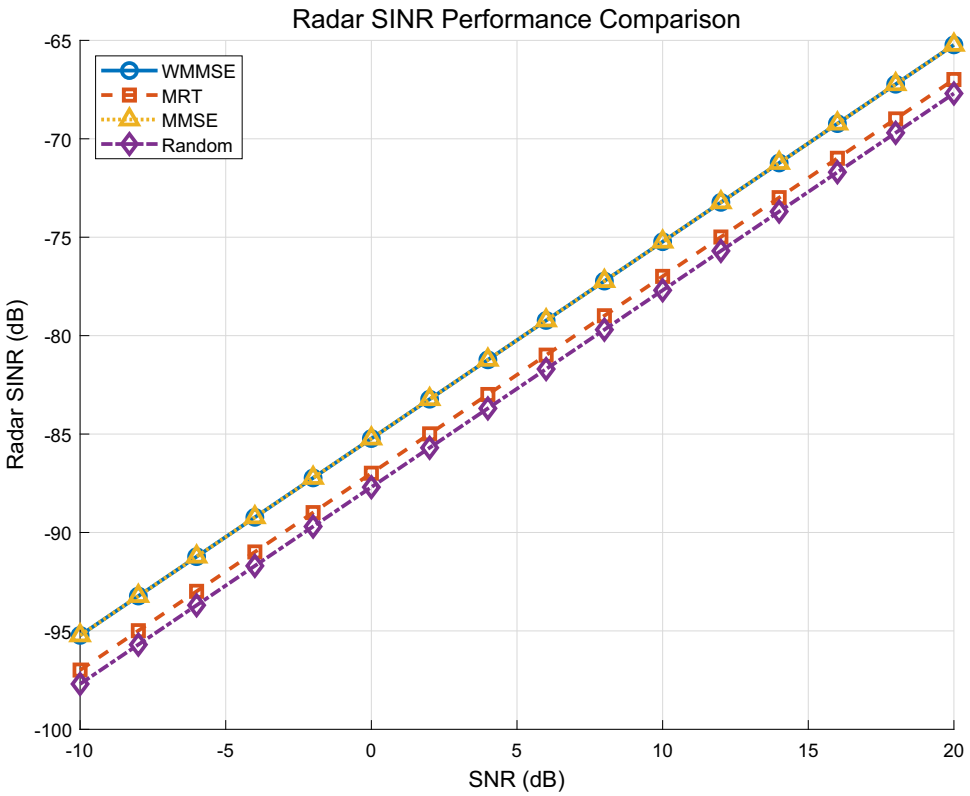


Fig. 4 Radar SINR versus SNR

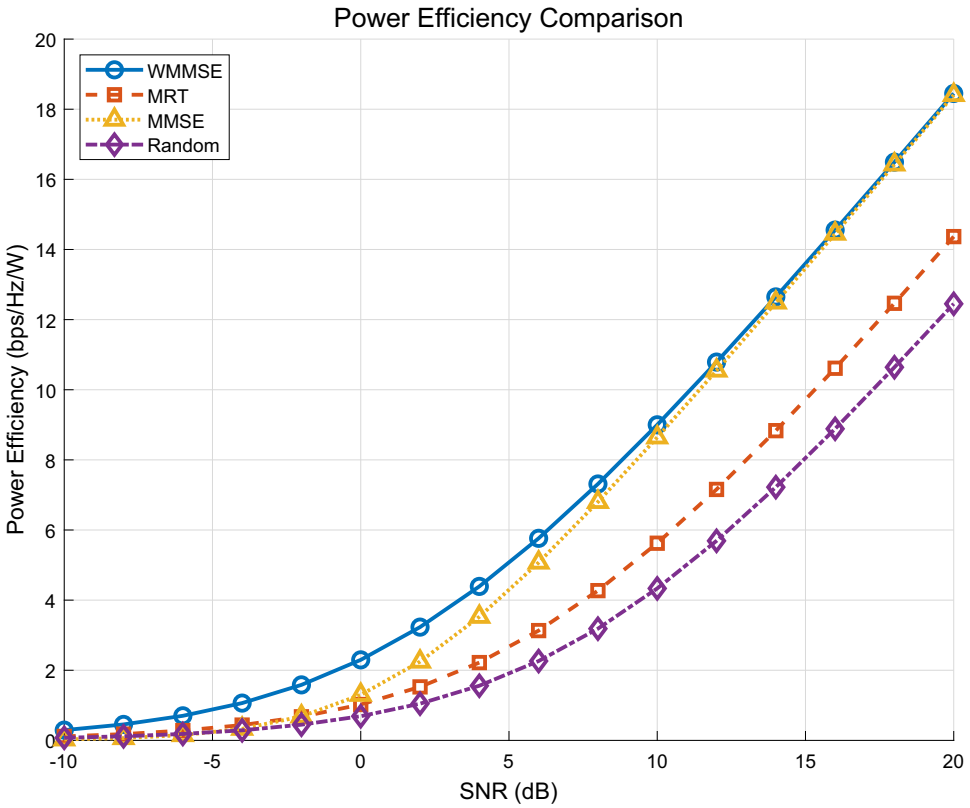


Fig. 5 System energy efficiency versus SNR

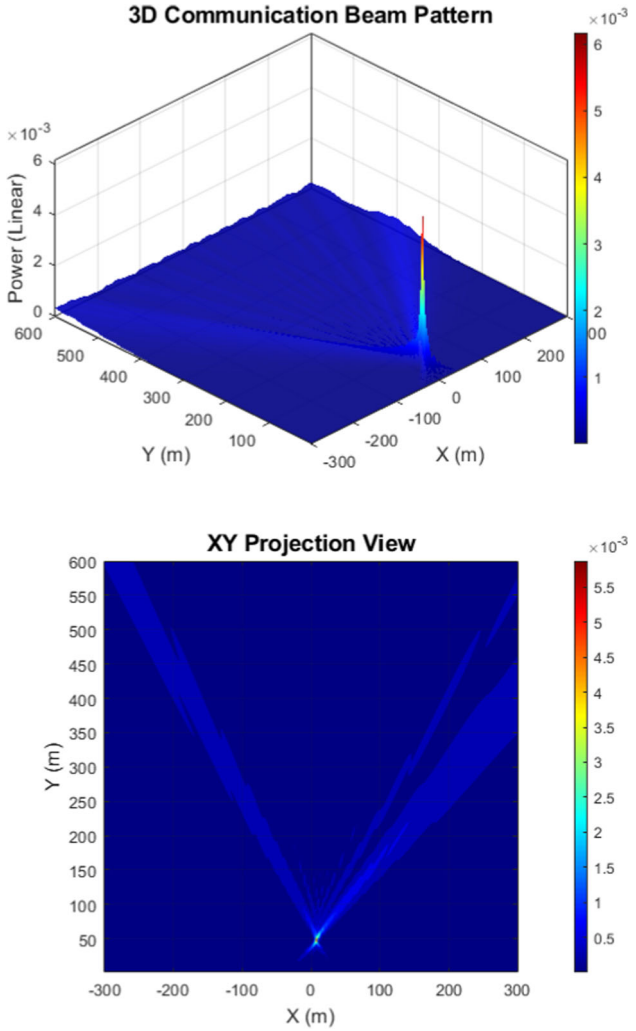


Fig. 6 Communication beam: near-field

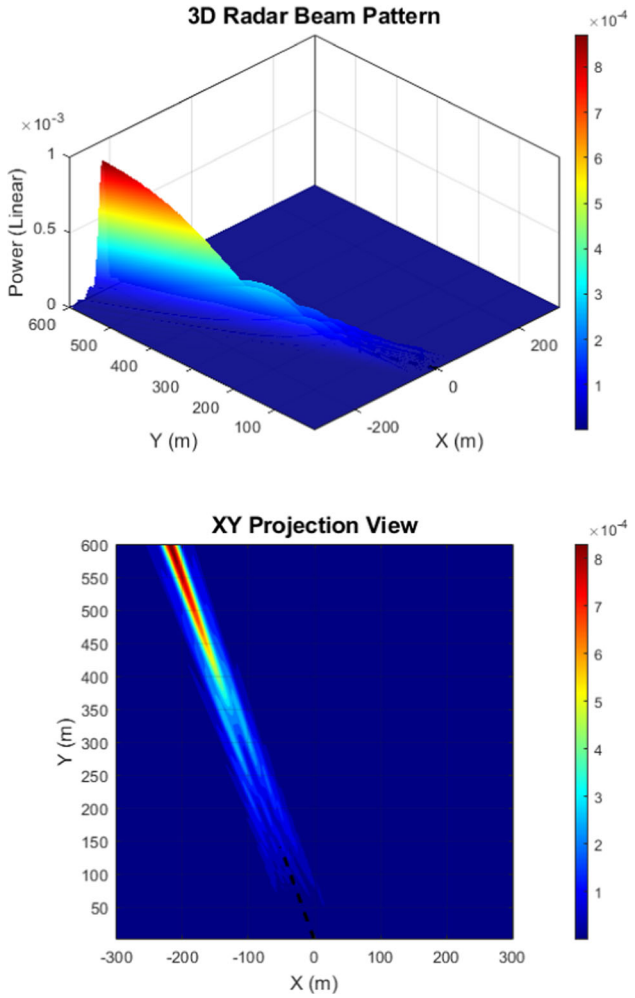


Fig. 7 Radar beam: far-field

Visualization of near-field and far-field beam patterns

To visualize the underlying mechanism, we simulate a single-user/single-target scenario with $N = 64$ at $f_c = 100\,\text{MHz}$, the user at $(r_c, \theta_c) = (50\text{m}, 10^\circ)$ and the target at $\theta_s = -20^\circ$.

Figure 6 shows the communication beam $\mathbf{w}_c$ focusing into a localized peak at (50m, 10°) rather than an angular ridge: phase-aligned spherical wavefronts produce constructive interference at that point and rapid roll-off elsewhere, quarantining the signal to a small near-field region and causing minimal leakage to the far-field radar.

Figure 7 shows the radar beam $\mathbf{w}_s$ as the classic far-field angular ridge along $\theta_s = -20^\circ$, range-invariant by design. In the near-field, the user at (50m, 10°) sits in a low, incoherent sidelobe portion: a far-field beam cannot focus tightly in the near-field, so its leakage onto the user is naturally small.

Overall, the simulations validate the proposed framework across sum-rate, per-user QoS, radar SINR feasibility, and energy efficiency, by turning the physical near/far-field distinction into a controllable design degree of freedom.

Conclusion

This paper introduced a range-domain interference-suppression paradigm for mixed near/far-field ISAC, exploiting the asymmetry between range–angle near-field channels and angle-only far-field sensing. A bi-directional mixed-field SINR model with range-dependent steering vectors was formulated, and a closed-form single-user analysis revealed that wavefront-curvature mismatch yields an interference-suppression gain even under full angular overlap, opening a degree of freedom unavailable in pure far- or near-field designs. The resulting non-convex sum-rate problem was solved by a WMMSE–SCA procedure with SOCP inner subproblems and guaranteed KKT-point convergence. Theoretical analysis and simulations confirmed that the proposed framework consistently outperforms representative baselines

while strictly meeting per-user and sensing constraints, providing a principled route toward jointly high-throughput and reliable sensing in future ISAC networks.

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Data Availability Not applicable

Materials Availability Not applicable

Code Availability Not applicable

Declarations

Conflict of interest Not applicable.

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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References

1. Liu F, Cui Y, Masouros C et al (2022) Integrated sensing and communications: toward dual-functional wireless networks for 6g and beyond. IEEE J Sel Areas Commun 40(6):1728–1767
2. Singh S, Samal U (2025) Integrated sensing and communication in next-generation wireless networks: insights and trends. Int J Commun Syst 38(5):70014
3. Kumari P, Choi J, González-Prelcic N, Heath RW (2018) Ieee 802.11ad-based radar: an approach to joint vehicular communication-radar system. IEEE Trans Veh Technol 67(4):3012–3027
4. Zheng L, Lops M, Eldar YC, Wang X (2019) Radar and communication coexistence: an overview: a review of recent methods. IEEE Signal Process Mag 36(5):85–99
5. Liu T, Guo Y, Lu L et al (2022) Waveform design for integrated sensing and communication systems based on interference exploitation. J Commun Inf Netw 7(4):447–456
6. Björnson E, Hoydis J, Sanguinetti L (2017) Massive mimo networks: spectral, energy, and hardware efficiency. Found Trends Signal Process 11(3–4):154–655
7. Hassanien A, Amin MG, Zhang YD, Ahmad F (2016) Dual-function radar-communications: information embedding using sidelobe control and waveform diversity. IEEE Trans Signal Process 64(8):2168–2181
8. Liu F, Masouros C, Li A et al (2018) Mu-mimo communications with mimo radar: from co-existence to joint transmission. IEEE Trans Wirel Commun 17(4):2755–2770
9. Xie C, Li Y, Chen L et al (2024) Power minimization for integrated sensing, communication, and power transmission systems. IEEE Commun Lett 28(12):2779–2783
10. Chu J, Liu R, Li M et al (2023) Joint secure transmit beamforming designs for integrated sensing and communication systems. IEEE Trans Veh Technol 72(4):4778–4791
11. Keskin MF, Wymeersch H, Koivunen V (2021) Mimo-ofdm joint radar-communications: is ici friend or foe? IEEE J Sel Top Signal Process 15(6):1393–1408
12. Ren Z, Qiu L, Xu J et al (2023) Robust transmit beamforming for secure integrated sensing and communication. IEEE Trans Commun 71(9):5549–5564
13. Zhu M, Li L, Xia S et al (2023) Information and sensing beamforming optimization for multi-user multi-target mimo isac systems. EURASIP J Adv Signal Process 2023(1):15
14. Tervo O, Tran LN, Juntti M (2015) Optimal energy-efficient transmit beamforming for multi-user miso downlink. IEEE Trans Signal Process 63(20):5574–5588
15. Chen K, Mao Y, Yin L et al (2024) Rate-splitting multiple access for simultaneous multi-user communication and multi-target sensing. IEEE Trans Veh Technol 73(9):13909–13914
16. Zhang Z, Liu Y, Wang Z et al (2024) Physical layer security in near-field communications. IEEE Trans Veh Technol 73(7):10761–10766
17. Cui M, Wu Z, Lu Y et al (2023) Near-field mimo communications for 6g: fundamentals, challenges, potentials, and future directions. IEEE Commun Mag 61(1):40–46
18. Zhang H, Shlezinger N, Guidi F et al (2022) Beam focusing for near-field multiuser mimo communications. IEEE Trans Wirel Commun 21(9):7476–7490
19. Hu S, Wang H, Ilter MC (2024) Design of near-field beamforming for large intelligent surfaces. IEEE Trans Wirel Commun 23(1):762–774
20. Wang Z, Mu X, Liu Y (2023) Near-field integrated sensing and communications. IEEE Commun Lett 27(8):2048–2052
21. Liu Y, Wang Z, Xu J, Ouyang C, Mu X, Schober R (2023) Near-field communications: a tutorial review. IEEE Open J Commun Soc 4:1999–2049

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